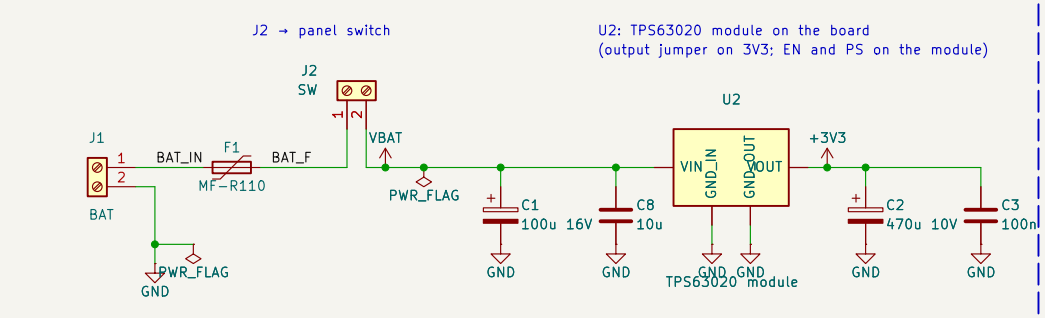
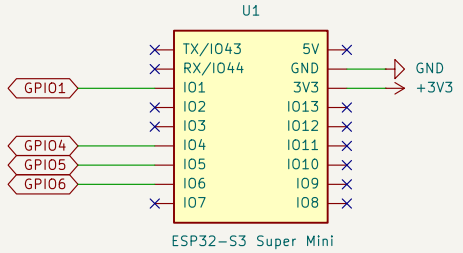
1 · BATTERY, SWITCH AND 3.3 V REGULATOR



F1 protects the whole board. C1/C8: U2 input.
C2/C3 next to the 3V3 pin of U1 (WiFi peaks of ~0.45 A).
TPS63020: VIN 1.8-5.5 V. With the scale charger connected, BAT must stay < 5.5 V.

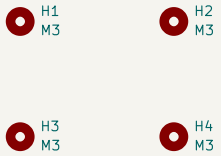
2 · ESP32-S3 SUPER MINI (IN SOCKET)



GPIO1 = SL, GPIO4 = DA (bus read only).
GPIO5 / GPIO6 = fans 1 and 2 (active high).

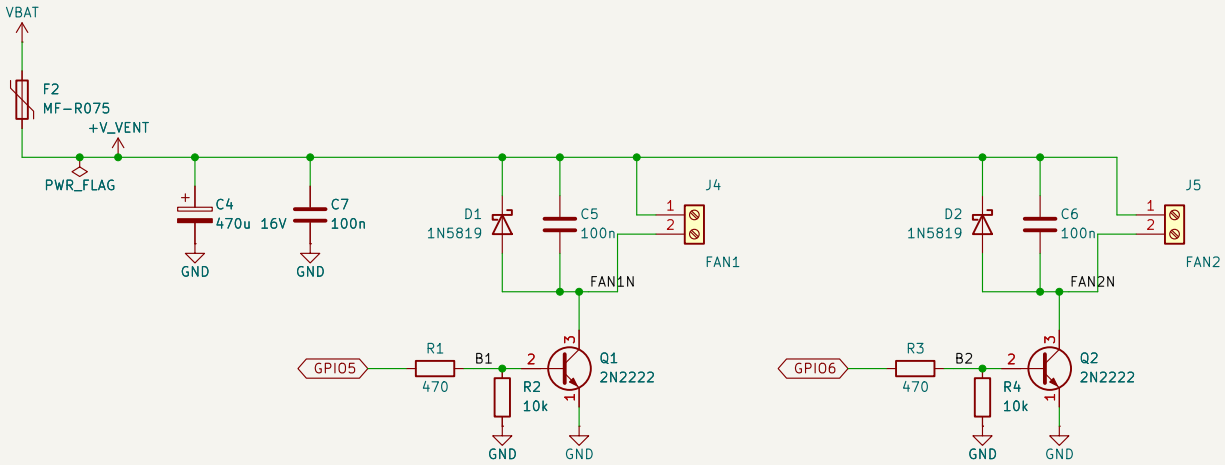
Powered through 3V3 only; the 5V pin is not connected.
No USB to the PC while the battery is switched on.

5 · MECHANICAL

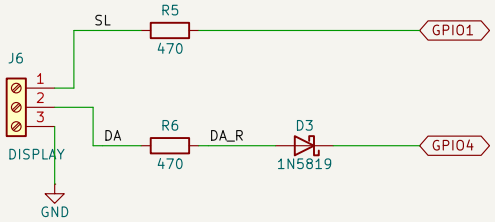


M3 holes (3.2 mm)
Board 96 × 53 mm
2 layers, FR-4 1.6 mm
1 oz copper, HASL

4 · 5 V FANS (LOW-SIDE SWITCHED)



3 · PASSIVE DISPLAY READING (TM1640)



R5/R6 (470 Ω) limit the current if a GPIO
is configured as an output by mistake.

D3 (Schottky, band toward the ESP) keeps the
ESP from injecting current into DA.

3.3 V bus: straight to the GPIO, no level shifting.

NOTES

1. J1: scale Li-ion 1S battery (≤ 4.2 V). J2: panel switch ≥ 2 A. U2: TPS63020 module with the output jumper on 3V3.
2. J4/J5: 5 V 0.16 A fan: red → 1 (+), black → 2 (−); yellow (tachometer) and blue (PWM) insulated.
3. J6: display connector (SL, DA, GND); the display VCC is not connected.
4. U1 in a 2 × 1x9 female socket, rows 15.24 mm apart; U2 on 4 pins at its corners. Q1/Q2 2N2222 E-B-C.
5. Diodes: band = cathode (square pad). Electrolytics: long lead to +.
6. PCB: tracks at 0/45/90°; power 1.0 mm, 3V3 and fans 0.8 mm, signal 0.3 mm; GND on both layers.
7. First power-up without U1 (with U2): measure 3.3 V between the 3V3 and GND pins of the U1 socket.

Verified: KiCad ERC and schematic-PCB parity
Generated by gen_sch.py; board by gen_pcb_v2.py
TM1640 sniffer and control of 2 fans; board 96 x 53 mm, TPS63020 on the board

Sheet: /		
File: caja_balanza_v2.kicad_sch		
Title: Scale box ESP32-S3		
Size: A4	Date: 2026-10-29	**Rev: 2.0**
KiCad E.D.A. 10.0.1		Id: 1/1